TULLY.

Autonomous workforce for fruit farms

We build and operate an autonomous robotic service that harvests, thins and takes on other orchard tasks. We start with the biggest market: apples.

Raising 3.5 M€ to prove the robot in the field



Labour is scarce and costly

LABOUR IS COSTLY

51%

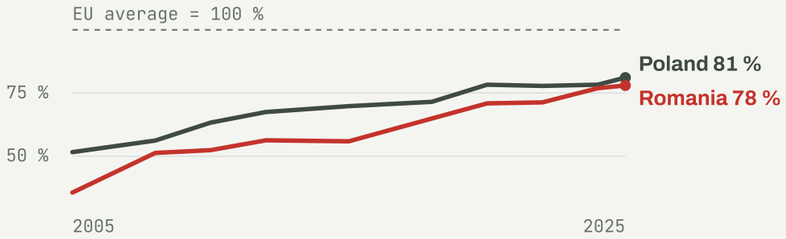
of an apple orchard's yearly running costs is hand labour: pruning, thinning, picking



Spraying, fertiliser 17 %
Machinery, fuel 16 %
Irrigation 9 %
Other 7 %

SEASONAL WORKERS ARE SCARCE

Their home economies are catching up, so fewer come each season



GDP per head in the main home countries of seasonal workers, % of the EU average

−27 % seasonal workers on German farms, 2010 to 2023



Fruit farm work, automated

Tully builds an autonomous robotic platform for the field work in orchards, starting with thinning and harvesting.

Base + Arms = THINNING

Base + Arms + Fruit line, bin = HARVESTING

Robotic system

Cameras find every apple, the hand cuts the stem, day and night

Modular

One base, a kit for each job in the orchard

Sold as a service

Farmers book the work, not the machine



TULLY H2 COBOT HARVESTER · RENDER

Founders who have built robots and companies before



Daniel Sattel

Co-founder & CEO

Second-time founder; scaled an IoT mobility company to seven-figure revenue

AWAKE Mobility · KONUX · dyson



Harsh Munshi

Co-founder & CTO

15+ years in computer vision, including perception for autonomous systems

KOPERNIKUS AUTOMOTIVE · NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE



Raphael Joppi

Co-founder & CPO

Takes robots from concept to deployment; grew up in an apple-farming family

KINEXON · FRANKA EMIKA

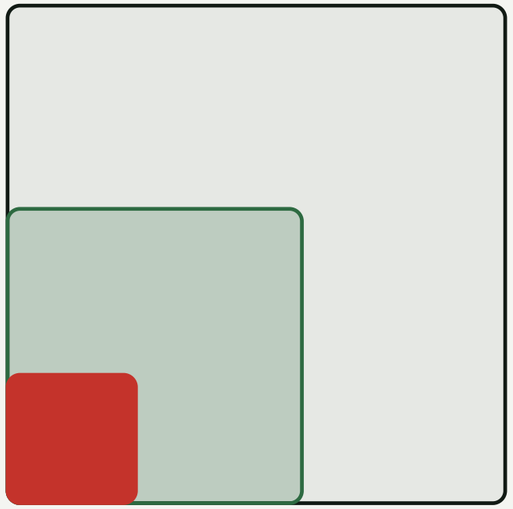
ADVISOR



Dr. Ines Hanrahan

Executive Director of the Washington Tree Fruit Research Commission; ten years in tree-fruit robotics research and grower needs.

Orchards spend 60 bn € a year on manual work



TAM **60 bn €**
orchards worldwide

SAM **21 bn €**
apples worldwide

SOM **4 bn €**
apples in 9 countries of western Europe and North America

462,000 ha
of apple orchards in the obtainable market

236 M
manual working hours a year

650–700 h
of hand work per hectare, each season

68 %
of US apples grow in Washington

Sold as a service

SERVICE

A fleet busy all season
More revenue and margin than selling each robot once

Fits how farms work
Booked like a seasonal crew

Nothing to buy
No upfront investment for the farmer

CUSTOMER

The farmer

The one who pays for seasonal crews today. Tully sends robots instead of a crew.

HUBS

Orchards cluster
Climate and terroir pack fruit farms into dense regions

One hub per region
Storage, dispatch, maintenance, live fleet monitoring

Low cost to serve
Many customers close by: little overhead, fast help in the field

Each robot pays for itself in 1.3 seasons

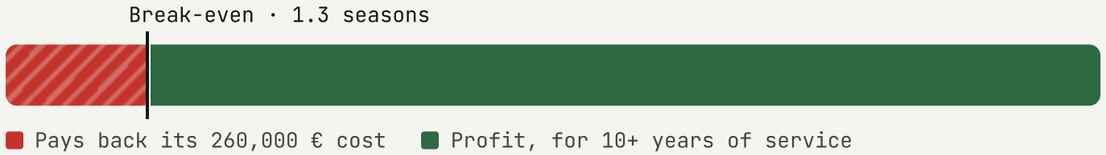
ONE ROBOT, ONE SEASON

227,909 €
revenue

195,658 €
gross profit

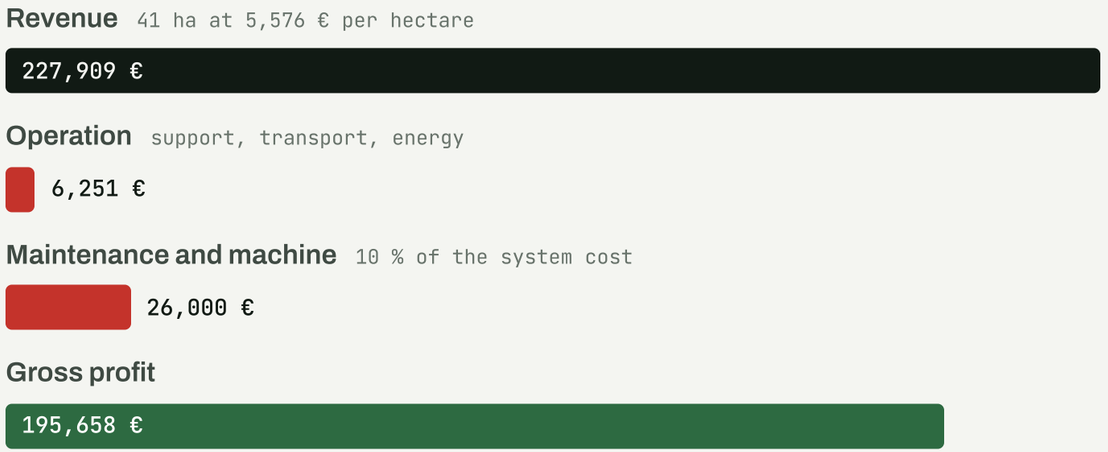
86 %
gross margin

≈1.7 M€
gross profit over 10 seasons, after the robot's cost



per apple	a day	per hectare	for the farmer
0.7 s	**20 h**	**3.3 d**	**−10 %**
vs 0.9 s on an assisted platform	vs 10 h	vs 7.4 days	cost per hectare

WHERE A SEASON'S REVENUE GOES



3.5 M€ to prove the robot in the field

24 months and two growing seasons.

01 **The first autonomous orchard robot with proven thinning and harvesting performance**

02 **The first services in commercial trials with farmers**

USE OF FUNDS





SIMULATION · THE MACHINE AT WORK

Commercial use in 2 years

THIS ROUND

YEAR 1
Working prototype: finds the apples and cuts them

YEAR 2
Robust machine with pilot customers: thinning, industrial apples

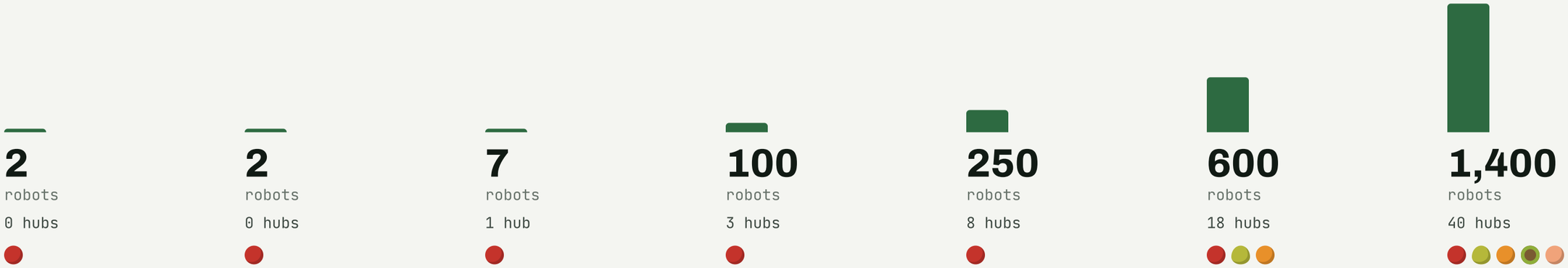
YEAR 3 COMMERCIAL
Commercial thinning and industrial apples from the first hub

YEAR 4
Thinning to harvesting in Washington, Bodensee, South Tyrol

YEAR 5
Into every leading apple region

YEAR 6
New fruits, global scale

YEAR 7
More fruits, global scale



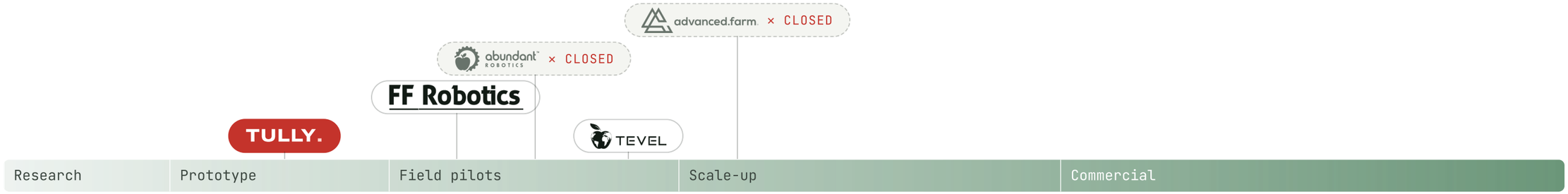
01
Thinning robot
From year 2 · ≈ 0.8 bn € a year · the shortest way to market

02
Harvesting system
From year 4 · ≈ 2.2 bn € a year · the same base plus fruit line and bins

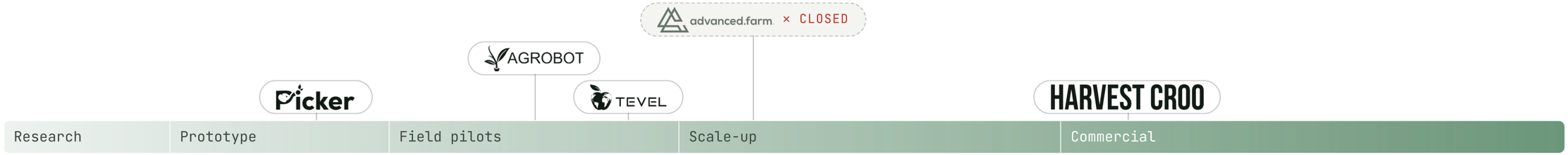
03
Fleets across orchards
From year 5 · every orchard job, region by region

No apple robot has reached commercial use yet

APPLES, THE LARGEST MARKET



OTHER FRUITS



WHY THEY FAILED, AND WHAT TULLY DOES

~~Built from scratch~~ → Off-the-shelf arms, platforms and cameras

~~Too long to market~~ → Thinning first, from year 2

~~Weak vision~~ → Today's AI models, depth cameras and edge compute

~~Selling the robot~~ → Sold as a service; Tully carries the risk

Farmers are ready to use Tully

30+ CONVERSATIONS WITH FARMERS

> *"Even if you can only automate one part of my manual work, the machine would run right now in my fields."*
>
> GROWER, INTERVIEWED BY TULLY

On demand

Available when the fruit is ready



0 €

No upfront investment

−10 %

Cheaper per hectare than picking by hand on a platform

0 h

No extra time to organise or supervise

~~Recruiting~~ ~~Housing~~ ~~Transport~~ ~~Paperwork~~ ~~Supervision~~

TULLY.

Autonomous workforce for fruit farms

Raising 3.5 M€ over 24 months and two growing seasons to prove the robot in the field, then roll out region by region, hub by hub.



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